Binomial random variables

1

a Independent

b Two possible outcomes: success and failure.

2

a Yes | b No | c Yes | d Yes

e Yes | f No

3

a The probability of success. | b The probability of failure.

4

a 0.3 | b 6

c The probability of exactly 2 successes.

5

a $\binom{4}{0}p^0q^4 + \binom{4}{1}p^1q^3 + \binom{4}{2}p^2q^2 + \binom{4}{3}p^3q^1 + \binom{4}{4}p^4q^0$

b The probability of exactly 2 successes.

Distributions of binomial random variables

6

a $p = 0.5$ | b $p = 0.15$ | c $p = 0.85$

7

a Equal | b Symmetric

8

a Greater than | b Positively skewed

9

a Less than | b Negatively skewed

10

a

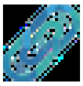
i Symmetric | ii $p = 0.50$

b

i Positively skewed | ii $p = 0.21$

c

i Negatively skewed



ii $p = 0.80$

Probability for binomial random variables

11

a 0.00046

b 0.94208

12

a $P(X = 2) = \binom{13}{2} \times (0.4)^2 \times (0.6)^{11}$

b 0.0453

13

a 0.0425

b $n = 10$

c $p = 0.6$

14

a $n = 4$

b $p = 0.4$

c ${}^4C_2 \times (0.4)^2 \times (0.6)^{4-2}$

d 6

e $P(X = 2) = 0.3456$

15

x	0	1	2	3
$P(X = x)$	0.343	0.441	0.189	0.027

16

a ${}^4C_1 \times (0.1)^1 \times (0.9)^{4-1}$

b ${}^4C_2 \times (0.1)^2 \times (0.9)^{4-2}$

c

x	0	1	2	3	4
$P(X = x)$	0.6561	0.2916	0.0486	0.0036	0.0001

17

a $n = 20$

b $p = 0.75$

c $P(X = 11) = 0.0271$

18

a $n = 6$

b $1 - p = 0.2$

c $\dfrac{256}{3125}$

d $\dfrac{309}{3125}$

a $\frac{992}{3125}$

b $\frac{2133}{3125}$

20

a 5

b 0.2

c $P(X > 3) = \frac{21}{3125}$

d 1

e 1

f 0.8

21

a 8

b $\frac{4}{5}$

c 0.94

d 7

e $\frac{32}{5}$

f $\frac{32}{25}$

22

a 0.52822

b 0.16308

23

a 4

b 0.0756

c 0.9163

d $p = 0.3$

e 1.2

f $\frac{\sqrt{21}}{5}$

24

a 5

b 0.31744

c 0.68256

d 0.07776

e $p = 0.6$

f $\sqrt{\frac{6}{5}}$

25

a 4

b 0.0256

c 0.9728

d $p = 0.2$

e 0.8

f $\frac{4}{5}$

26

a 5

b 0.83692

c 0.16308

d 0.00243

e $p = 0.3$

f $\sqrt{\frac{21}{20}}$

27

a 0.07

b 0.28

c 0.75

d 0.41

e 0.22

28

a $p = 0.6$ **b** 2.4 **c** 0.96

29

a $p = \frac{3}{5}$ **b** $\frac{12}{5}$

30

a $n = 40$ **b** $p = 0.2$

31

a $n = 25$ **b** $\frac{\sqrt{21}}{2}$

32

a $p = 0.2$ **b** $n = 10$

33

a $p = 0.4$ **b** $n = 12$

Applications

34 70 people

35 65

36

a $\frac{1}{6}$ **b** 0.1608 **c** 0.0322

37

a $\frac{3}{11}$ **b** 0.2861 **c** 0.2035

38

a 0.2212 **b** 0.0125

39

a $\binom{5}{2} \times \left(\frac{1}{5}\right)^2 \times \left(\frac{4}{5}\right)^3$

b The probability of selecting exactly 3 milk chocolates.

c $\frac{256}{625}$

d $\frac{2304}{3125}$